Probability Exercises - Solutions

Exercise 1.1.

a) *A spinner has eight sections, each of which is equally likely to land on the table when the spinner is used. The sections are numbered 1-8. What is the probability that the spinner will land on a square number?*
Of the numbers 1-8, the square numbers are 1 and 4. This is two out of eight possible outcomes, all of which are equally likely, so the probability is $\frac{2}{8} = \frac{1}{4}$.

b) *A set of 100 Scrabble tiles, with the distribution shown, is placed in a bag, and one tile is drawn out. What is the probability it is a vowel?*
$12\times\boxed{E}$ $9\times\boxed{A}\boxed{I}$ $8\times\boxed{O}$ $6\times\boxed{N}\boxed{R}\boxed{T}$ $4\times\boxed{D}\boxed{L}\boxed{S}\boxed{U}$ $3\times\boxed{G}$ $2\times\boxed{B}\boxed{C}\boxed{F}\boxed{H}\boxed{M}\boxed{P}\boxed{V}\boxed{W}\boxed{Y}\boxed{}$ $1\times\boxed{J}\boxed{K}\boxed{Q}\boxed{X}\boxed{Z}$
Of the 100 tiles, there are 9 $\boxed{A}$s, 12 $\boxed{E}$s, 9 $\boxed{I}$s, 8 $\boxed{O}$s and 4 $\boxed{U}$s, which is a total of 42 vowels. This is 42 out of 100 possible outcomes, all of which are equally likely, so the probability is $\frac{42}{100} = \frac{21}{50}$.

c) *In the UK lottery, six balls are drawn from a set numbered 1-59 in the machine, which is carefully designed to ensure each ball has an equal probability of being drawn. What is the probability that the first ball drawn is an odd number?* Of the 59 balls in the draw, 30 of them are odd numbers (including 1 and 59). This is 30 out of 59 possible outcomes, all of which are equally likely, so the probability is $\frac{30}{59}$.

d) *A lecturer surveyed the 29 students in their class about how they travelled to university. 5 students walked, 7 students cycled, 6 students travelled by bus, and the rest of the students in the class travelled by car. A student is picked at random from the class. Find the probability that the student travelled to university either by bus or by car.*
Of the 29 students, 6 travelled by bus, and $29 - (5 + 7 + 6) = 11$ travelled by car. This is 11 out of 29 possible outcomes, all of which are equally likely since the student is picked at random, so the probability is $\frac{11}{29}$.

Exercise 1.2. *Are the following events dependent or independent?*

a) *Ali spins a spinner once a day for a week. $A =$ 'on Tuesday, it lands on 1'; $B =$ 'on Friday, it lands on 3'*
Independent

b) *A playing card is drawn from a deck of cards. $A =$'the card is a club'; $B =$ 'the card is a 7'.* Independent

c) *A playing card is drawn from a deck of cards. $A =$'the card is a club'; $B =$ 'the card is black'.* Dependent

d) *A random MMU student is chosen. $A =$ 'the student is taller than the median height'; $B =$ 'the student is a member of the MMU basketball team'* Dependent

Exercise 1.3. *Which of the following are mutually exclusive?*

a) *A playing card drawn from a deck is red; it is black* Mutually exclusive

b) *A playing card drawn from a deck is red; it is a club* Mutually exclusive

c) *A playing card drawn from a deck is red; it is a diamond* Not mutually exclusive

d) *A die is rolled and the result is a 6; the result is a 5* Mutually exclusive

e) *A die is rolled and the result is a 6; the result is odd* Mutually exclusive *f) The maximum temperature tomorrow will be $5°c$; the minimum temperature tomorrow will be $6°c$* Mutually exclusive

g) A participant in a study has never eaten chips; they eat chips at least once a week Mutually exclusive

h) A participant in a study has never eaten chips; they have never been to the gym Not mutually exclusive

Exercise 1.4.

The probabilities of various events are given, regarding a normal UK Lottery draw.

$$A = \text{'all the balls drawn are numbers greater than 10'} \qquad P(A) = 31\%$$

$$B = \text{'the ball set "Guinevere" is chosen for the draw'} \qquad P(B) = 25\%$$

$$C = \text{'the jackpot for the draw is more than £10m'} \qquad P(A) = 17\%$$

$$D = \text{'more than the median number of tickets are sold'} \qquad P(A) = 50\%$$

For each of the joint probabilities listed below, assuming the events are independent, calculate the joint probability. Discuss whether all of these are independent, and what information you would need in order to determine this.

$$i)\ P(A \cap B) \qquad ii)\ P(A \cap C) \qquad iii)\ P(B \cap D) \qquad iv)\ P(C \cap D)$$

(i) $P(A \cap B) = 0.31 \times 0.25 = 0.0775 = 7.75\%$
(ii) $P(A \cap c) = 0.31 \times 0.17 = 0.0527 = 5.27\%$
(iii) $P(B \cap D) = 0.25 \times 0.50 = 0.125 = 12.5\%$
(iv) $P(C \cap D) = 0.17 \times 0.5 = 0.085 = 8.5\%$

Events A and B are independent from the others, as they are random draws conducted under controlled conditions (the ball set is chosen from four possible options, at random before each draw). The size of the jackpot is calculated based on the number of tickets sold, so we would expect C and D to be dependent.

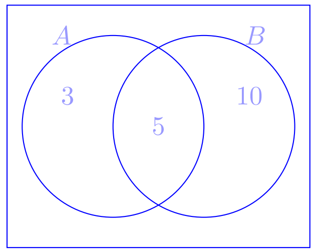
Exercise 1.5.

i) In a class of 30 students, 8 students are in K-Pop Society, 15 students play a sport, and 5 students are both in the K-Pop Society and play a sport. Let A be the event that a student is in the K-Pop Society and let B be the event that a student plays a sport. Find $P(A \cap B)$, and find $P(A \cup B)$.
By counting, $P(A \cap B) = \frac{5}{30} = \frac{1}{6}$.
$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{8}{30} + \frac{15}{30} - \frac{5}{30} = \frac{8+15-5}{30} = \frac{18}{30} = \frac{3}{5}$.

ii) Nic sometimes has a dessert after dinner. Nic has a cookie for dessert on average one night in every four, and ice cream for dessert one night in 10. Sometimes, when it's a special occasion (one night in 20), dessert is both a cookie and ice cream. What is the probability that Nic will have dessert tonight?
Let $C =$ 'Nic has a cookie tonight' and $I =$ 'Nic has ice cream tonight'. Then $P(C) = \frac{1}{4}$, and $P(I) = \frac{1}{10}$. We are told that $P(I \cap C) = \frac{1}{20}$. Then the probability of having a dessert of any kind is $P(I \cup C) = P(I) + P(C) - P(I \cap C) = \frac{1}{10} + \frac{1}{4} - \frac{1}{20} = \frac{2+5-1}{20} = \frac{6}{20} = \frac{3}{10}$.

Exercise 1.6.

a) *In a class of 30 students, 8 students are in K-Pop Society, 15 students play a sport, and 5 students are both in the K-Pop Society and play a sport. Let A be the event that a student in the class is in the K-Pop Society and let B be the event that a student in the class plays a sport. Draw a Venn diagram to illustrate this. State $P(A), P(B), P(A|B)$ and $P(B|A)$, and use Bayes' Theorem to check they are related correctly.*



$P(A) = \frac{8}{30}$ and $P(B) = \frac{15}{30}$. By counting, $P(A|B) = \frac{5}{15} = \frac{1}{3}$ (since of the 15 students who play a sport, 5 of them are also in K-Pop society) and similarly $P(B|A) = \frac{5}{8}$.

Bayes' theorem states that:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} = \frac{\frac{5}{8} \times \frac{8}{30}}{\frac{15}{30}} = \frac{\frac{5}{30}}{\frac{15}{30}} = \frac{1}{3}, \textit{ as required.}$$

b) *A poorly child comes to the doctor, who treats only children with the flu and measles. 90% of poorly children in the area have the flu, while the other 10% have measles. Let F = 'a child has flu' and M = 'a child has measles'.*

A well-known symptom of measles is a rash, and let R = 'a child has a rash'. Let the probability of having a rash if you have measles be $P(R|M) = 0.95$. A small number of children with the flu also develop a rash, and $P(R|F) = 0.08$.

The doctor examines a child, and finds they have a rash. What is the probability that the child has measles?
We know that $P(F) = 90\% = 0.9$ and $P(M) = 10\% = 0.1$.

The probability that a child has a rash will be a combination of the probability of rash due to flu, and rash due to measles: if they have flu, they have an 8% chance of rash (or, $P(R|F) = 0.08$), and if they have measles there is a 95% chance of a rash (or, $P(R|M) = 0.95$).

So $P(R) = P(F)P(R|F) + P(M)P(R|M) = (0.9 \times 0.08) + (0.1 \times 0.95) = 0.072 + 0.095 = 0.167$. Then, by Bayes' theorem, the probability that a child with a rash has flu is:

$$P(F|R) = \frac{P(R|F)P(F)}{P(R)} = \frac{0.95 \times 0.1}{0.167} = 0.568 = 57\%$$

So, even though rash is strongly associated with measles, since flu is so prevalent, it's more likely you have flu than measles if you present with a rash.

Exercise 1.7. *Given the example table of probabilities above, write down a function that would give the probability of the result being 2 or 3, and write down a function to give the probability of the result being 6.*

=PROB(B3:B8,C3:C8,2,3) *=PROB(B3:B8,C3:C8,6)*